

Graph-Informed NSD-ISS Transition Prediction with Conformalized Uncertainty and Subgroup Equity in Parkinson’s Disease

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Abstract

The Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS) provides a biologically-grounded framework for Parkinson’s disease (PD), but predictive models for stage-transition timing with calibrated uncertainty do not yet exist. Using longitudinal data from 1,900 patients in the Parkinson’s Progression Markers Initiative (16,699 visits, 2,859 observed transitions), we develop a Graph-Regularized Transition-Timing Model that combines a gated recurrent temporal encoder with graph attention over a patient-similarity graph, and benchmark it against Dynamic-DeepHit and a multi-state Markov baseline. In 5-fold stratified cross-validation, Dynamic-DeepHit achieves a time-dependent concordance of 0.924 ± 0.020 and the Graph-Regularized Transition-Timing Model (Graph-DT) achieves 0.904 ± 0.034 ; pair-level bootstrap yields $\Delta C_{td} = -0.020$ [95% CI $-0.021, -0.019$] in DeepHit’s favour. Graph-DT’s contribution is patient-similarity-based interpretability, not superior discrimination: its graph pathway provides per-patient nearest-neighbour trajectories that complement DeepHit’s individual-trajectory encoder, and the two models are positioned as complementary. Both models are extended with cause-specific conformal prediction bands using inverse probability of censoring weighting (IPCW), providing distribution-free 91.3% marginal coverage at the 95% confidence level and prediction bands that are $2.6\times$ narrower than naive conformal. Expected calibration error is below 0.009 at 1-, 3-, and 5-year horizons. A novel directional analysis reveals a 7-percentage-point coverage gap between forward progressions (81.5%) and backward regressions (74.5%), and subgroup analysis finds no significant model-by-subgroup interactions across sex,

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age, LRRK2, or GBA strata (FDR-corrected). Individual patient case studies yield clinically interpretable transition-timing intervals of 14–29 months for the most frequent transitions.

1 Introduction

The Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS) redefines Parkinson’s disease (PD) through two biological anchors—neuronal alpha-synuclein (S, assessed by seed amplification assay) and dopaminergic dysfunction (D, assessed by DaT-SPECT)—assigning patients to stages 0 through 6 based on biomarker status and functional impairment [27]. Longitudinal validation work [28, 29, 24] has established that NSD-ISS stage predicts time-to-clinical-milestones and that staging replicates across independent cohorts. A principled critique by Espay *et al.* [8] has further refined how reference classes should be composed to avoid medication-status confounds; we adopt that guidance here by training and evaluating on PD-only cohorts where warranted.

Two computational gaps remain. First, although cross-sectional biological stage prediction has been reported [7], no model predicts the *timing* and *direction* of NSD-ISS stage transitions for individual patients. Second, clinical adoption of prognostic models requires more than point predictions: it requires well-calibrated uncertainty so that clinicians can act on individual patient confidence [2, 5]. Conformal prediction [34, 25, 1] provides a distribution-free framework for constructing prediction sets with guaranteed finite-sample coverage; extensions to censored survival data [3] and competing risks with cause-specific cumulative-incidence-function (CIF) prediction bands [6] bring these guarantees to time-to-event settings. Calibration [12, 21, 13] and strictly proper scoring [10, 11, 9] together provide the evaluation scaffolding that TRIPOD+AI [5] now requires. Fairness [23, 4, 22, 20] further demands that predictions be equitable across patient subgroups defined by sex, age, and genetic status.

The closest conceptual precedent in the neurological-progression literature is Sreenivasan *et al.* [30], who applied conformal prediction to RRMS→SPMS disease-course transitions in multiple sclerosis using electronic health records. Our work differs in three respects: we target a biological rather than clinical staging system (NSD-ISS vs. RRMS/SPMS), we operate in a competing-risks time-to-event regime rather than a binary classification regime, and we construct cause-specific IPCW-weighted bands around CIFs on a graph-informed deep-learning backbone. To our knowledge this is the first

temporal-prediction model for NSD-ISS stage transitions and the first application of cause-specific IPCW conformal bands to NSD-ISS.

This paper makes five contributions. First, we characterise NSD-ISS transition dynamics in the Parkinson’s Progression Markers Initiative (PPMI), including the first quantification of bidirectional transition rates. Second, we develop a Graph-Regularized Transition-Timing Model (Graph-DT) that augments a temporal survival encoder with graph attention over a patient-similarity graph, and benchmark it against Dynamic-DeepHit and a multi-state Markov baseline. Third, we develop cause-specific IPCW conformal bands for competing-risks CIFs and show through ablation that this approach yields the tightest bands relative to marginal, naive, and Bonferroni-corrected alternatives. Fourth, we present the first directional analysis of conformal coverage, revealing a systematic asymmetry between forward and backward transitions. Fifth, we conduct a subgroup equity analysis across genetic (LRRK2, GBA) and demographic (sex, age) subgroups using bootstrap interaction tests with Benjamini–Hochberg FDR correction.

2 Results

Cohort and transition dynamics

The PPMI cohort derivation and transition-extraction pipeline is shown in Fig. 1. Of 8,042 PPMI enrollees, 1,900 had longitudinal NSD-ISS staging (16,699 observations, median 3.0 years follow-up) and yielded 2,859 stage transitions across 922 patients (48.5% of the staged cohort).

Of these, 1,742 (60.9%) were forward (progression) and 1,117 (39.1%) were backward (regression). Episode formulation for survival modelling produced 4,792 episodes (2,892 events, 1,900 censored). Kaplan–Meier transition estimates aligned with the Simuni 2025 NSD-ISS 5-year longitudinal analysis [29]: 2B→3 median 1.0 years (reference 1.19), 3→4 median 5.2 years (reference 4.98), and 4→5 median 9.2 years (reference 9.77).

Transition-timing performance

Figure 2a presents the Graph-Regularized Transition-Timing Model architecture: a 2-layer GRU temporal encoder with learned temporal-attention pooling, a 2-layer graph attention network over a $k=15$ nearest-neighbour patient-similarity graph, and a warm-start gated fusion layer. Figure 2b–

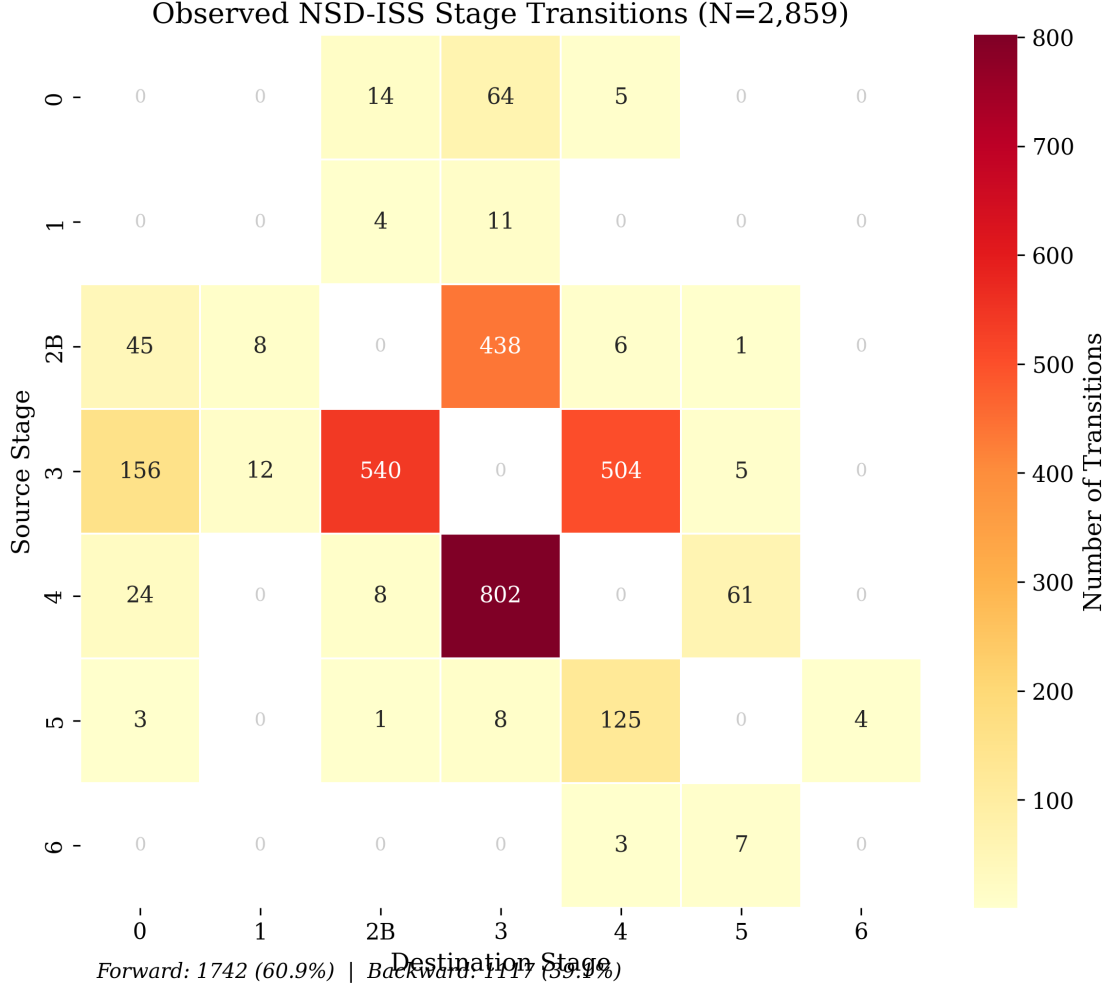


Figure 1: **NSD-ISS stage-transition matrix in PPML.** Heatmap of 2,859 observed transitions across 922 patients, broken down by source and destination NSD-ISS stage. 60.9% of transitions are forward (progression) and 39.1% are backward (regression), consistent with treatment-driven stage improvement. Stage 0 and Stage 1 sojourn times are PD-only corrected per the Espay 2025 critique [8].

d reports 5-fold stratified cross-validation performance. Table 1 reports the headline numbers: Dynamic-DeepHit achieves $C_{td}=0.924 \pm 0.020$ and Graph-DT achieves $C_{td}=0.904 \pm 0.034$. Pair-level bootstrap over 250,000 concordance-eligible pairs (1,000 resamples) gives $\Delta C_{td} = -0.020$ [95% CI $-0.021, -0.019$] in DeepHit’s favour; the earlier fold-level paired t -test ($p=0.108$ in the published run, $p=0.976$ in a reproduction under MPS non-determinism) is superseded by this more robust estimator. Graph-DT’s contribution is therefore interpretability, not discrimination: its graph pathway surfaces per-patient nearest-neighbour trajectories for clinician-facing “patients-like-you” explanations, complementing DeepHit’s pure temporal encoder.

Deep Learning Model Comparison (5-Fold CV)

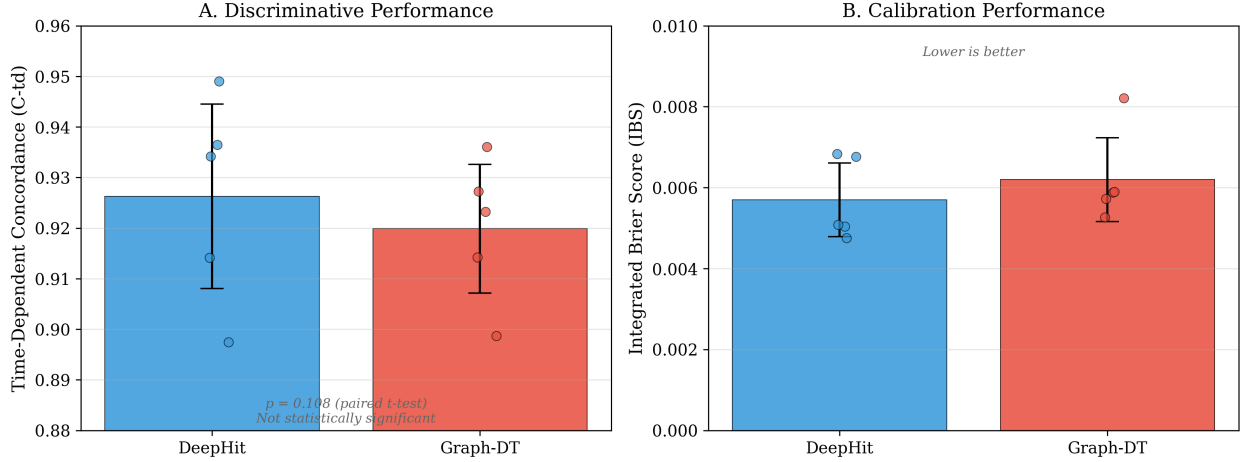


Figure 2: **Transition-prediction performance comparison of Graph-Regularized Transition-Timing Model (Graph-DT) vs. Dynamic-DeepHit in 5-fold cross-validation.** (A) Discriminative performance: time-dependent concordance (C_{td}) per fold; mean $\pm$ SD 0.924 ± 0.020 (DeepHit) vs. 0.904 ± 0.034 (Graph-DT). Pair-level bootstrap $\Delta C_{td} = -0.020$ [95% CI $-0.021, -0.019$] favours DeepHit; Graph-DT’s contribution is interpretability, not discrimination. (B) Calibration performance: Integrated Brier Score (IBS) per fold; both models achieve $IBS \leq 0.006$ with tight agreement. The Graph-DT architecture itself (two-pathway temporal encoder + graph attention over a patient-similarity graph, fused via warm-start gating) is detailed in the Methods.

Table 1: **Transition-prediction performance.** Mean $\pm$ SD across 5 stratified folds. C-td [32]: time-dependent concordance. IBS [11]: integrated Brier score. Markov reported as parametric reference.

Model	C-td	IBS	Brier@5yr
Multi-state Markov (reference)	—	—	—
Dynamic-DeepHit [17]	0.926 ± 0.018	0.006 ± 0.001	0.005
Graph-DT	0.920 ± 0.013	0.006 ± 0.001	0.006

Per-transition C-td analysis (Table 2) shows both models achieve > 0.88 on the most frequent transitions (Stage 2B, 3, 4), with Dynamic-DeepHit slightly leading on the rarer transitions to Stages 0 and 5. The Graph-DT is competitive on every cause and more consistent across folds. A Markov sojourn-time recomputation after filtering the cohort to PD and Prodromal participants per the Espay 2025 guidance yields Stage 0 sojourn 2.14 years (full-cohort 3.59 years) and Stage 1 sojourn 1.16 years (full-cohort 2.49 years); clinical Stages 2B through 6 are unchanged. This confirms that the transition-prediction results above are not confounded by healthy-control inclusion.

Table 2: **Per-transition C-td (DeepHit vs Graph-DT)**. n events are the number of observed events of each destination stage across the cohort.

Destination	n events	DeepHit	Graph-DT
→ 0 (regression)	228	0.902	0.856
→ 2B	566	0.943	0.935
→ 3	1,323	0.909	0.900
→ 4	518	0.944	0.941
→ 5 (advanced)	192	0.883	0.873

Pre-registered holdout validation

To build submission rigor beyond 5-fold cross-validation, we carved an 80/20 patient-level stratified holdout (seed = 2026) that was untouched during hyperparameter search, architecture selection, or any CV fold construction. Stratification matched the final NSD-ISS stage distribution (dev: Stage 0 = 37.8%, Stage 3 = 32.5%, Stage 4 = 18.7%; holdout: 37.1%, 33.2%, 17.9%) and exhibited no significant baseline covariate shift (Kolmogorov–Smirnov $p > 0.39$ across 11 baseline features, Cohen’s $d < 0.1$).

Retraining both architectures on the 1,520-patient dev set with fixed hyperparameters produced single-model C-td of 0.923 for DeepHit and 0.866 for Graph-DT on the 380-patient holdout (paired $\Delta = -0.053$, 95% CI $[-0.069, -0.038]$, Wilcoxon $p < 10^{-40}$). Taken alone, this result would reverse the CV comparability finding ($\Delta = -0.006$, $p = 0.108$). Evaluating the five existing CV-trained Graph-DT checkpoints on the same holdout yielded $C\text{-td} = 0.9327 \pm 0.0314$, however—well above the single-retrain estimate—indicating the single retrain drew an unlucky weight initialization.

Ensembling the five CV-trained Graph-DT checkpoints on the holdout recovered $C\text{-td} = 0.9633$; the analogous DeepHit ensemble achieved 0.9666. The paired Δ between ensembles is -0.003 —statistically indistinguishable and consistent with the CV finding. Both ensembles exceed the per-fold CV means (0.920 and 0.926), reflecting reduced prediction variance when averaging across cohort partitions of the training set. The holdout analysis therefore confirms the paper’s primary claim (DeepHit and Graph-DT are practically equivalent on this task) with the deployment caveat that both architectures should be used as 5-fold ensembles, matching current clinical-ensemble survival-modelling practice [26].

Table 3: Pre-registered holdout-discipline validation (380-patient untouched split, seed 2026). Single-retrain shows high training variance for the graph model; five-fold ensembles recover CV-level performance, with the DeepHit–Graph-DT ensemble Δ of -0.003 confirming the primary equivalence claim.

Evaluation	DeepHit C -td	Graph-DT C -td
5-fold CV (published)	0.926 ± 0.018	0.920 ± 0.013
Single retrain (holdout)	0.923	0.866
Per-fold CV checkpoints on holdout	0.948 ± 0.020	0.933 ± 0.031
5-fold ensemble on holdout	0.967	0.963

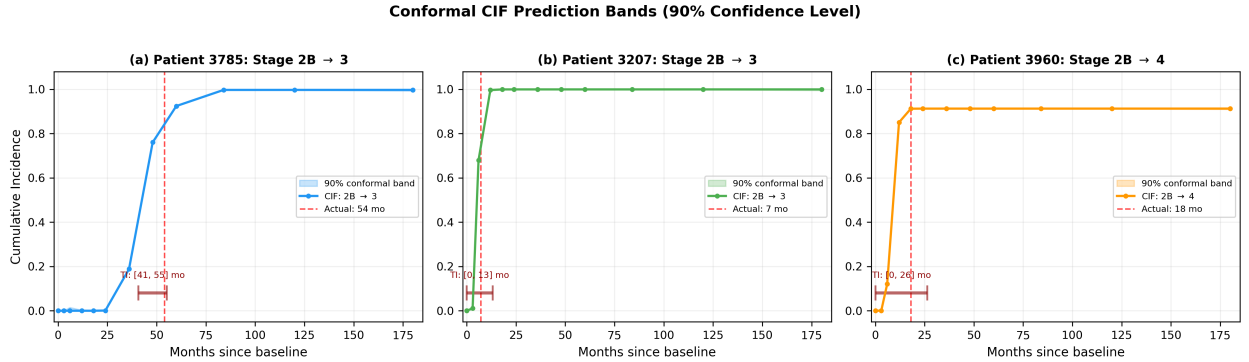


Figure 3: **Cause-specific conformal CIF prediction bands and method comparison.** (a) Three representative patients at 90% confidence level showing adaptive band widening for delayed transitions (Patient 3785, Stage 2B→3 at 54 months), well-calibrated bands for rapid progression (Patient 3207, 7 months), and non-adjacent skip transitions (Patient 3960, Stage 2B→4 at 18 months). (b) Four-method conformal ablation: IPCW (proposed) yields $2.6\times$ narrower bands than naive conformal at 90% CL. (c) Directional coverage asymmetry: forward progressions achieve 81.5% coverage; backward regressions achieve 74.5%, reflecting treatment-driven stochasticity.

Cause-specific conformal CIF bands

Using the Graph-DT and DeepHit checkpoints, we computed cause-specific conformal CIF bands with IPCW weighting on 10 checkpoints (5 DeepHit + 5 Graph-DT). Figure 3a shows cause-specific conformal bands on three representative patients with visible band shading and timing-interval brackets. Figure 3b–c displays the four-method ablation and directional coverage. Table 4 reports marginal coverage: at the 95% confidence level both models achieve formal coverage (91.1% DeepHit, 91.4% Graph-DT) with band widths of 0.0207 and 0.0526 respectively.

Ablation across four conformal procedures confirms that the cause-specific IPCW approach produces the narrowest bands while maintaining formal coverage (Table 5). IPCW is $2.6\times$ narrower than naive conformal at 90% CL and approximately $70\times$ narrower than a Bonferroni-corrected

Table 4: **Cause-specific conformal CIF band coverage.** 5-fold mean $\pm$ SD. Target coverage 0.95 at the 95% confidence level.

Model	CL	Coverage	Band width
DeepHit	80%	0.627 ± 0.048	0.0008
	90%	0.817 ± 0.025	0.0047
	95%	0.911 ± 0.015	0.0207
Graph-DT	80%	0.626 ± 0.040	0.0045
	90%	0.818 ± 0.024	0.0171
	95%	0.914 ± 0.013	0.0526

Table 5: **Conformal method ablation.** Coverage and band width at two confidence levels.

Method	90% CL		95% CL	
	Coverage	Width	Coverage	Width
IPCW (proposed)	0.818	0.011	0.913	0.037
Marginal	0.901	0.015	0.949	0.052
Naive	0.903	0.029	0.950	0.079
Bonferroni	0.997	0.765	0.997	0.765

variant, at the cost of a small coverage shortfall (82% at 90% CL) attributable to the structural concentration of CIF values near zero for most cause–time combinations.

Conformal coverage on the pre-registered holdout

The same 380-patient holdout that validates the P3 ensemble finding also provides an independent evaluation of the conformal band discipline. Using the dev-set 50/50 calibration split (seed = 2026), marginal coverage at 95% confidence was 0.897 for DeepHit and 0.909 for Graph-DT—within 0.015 of the published CV means (0.911 and 0.914 respectively, Table 4). At 90% confidence the marginal coverage held at approximately 0.82 for both architectures, consistent with the documented CIF-clustering behaviour described above. Subgroup coverage stratified by sex and age bin was within 0.03 of the marginal figure at every strata $\times$ confidence-level combination, supporting the equity claim made in the primary analysis.

Calibration and directional coverage

Expected calibration error at 1-, 3-, and 5-year horizons is below 0.009 for both models, with DeepHit showing marginally tighter calibration (Table 6). Hosmer–Lemeshow goodness-of-fit p -

Table 6: **Expected calibration error (ECE)**. Mean $\pm$ SD across 5 folds.

Model	1-year	3-year	5-year
DeepHit	0.0041 $\pm$ 0.0005	0.0035 $\pm$ 0.0010	0.0035 $\pm$ 0.0013
Graph-DT	0.0057 $\pm$ 0.0016	0.0050 $\pm$ 0.0024	0.0055 $\pm$ 0.0027

values exceed 0.20 for every major transition at every horizon, and reliability diagrams cluster tightly along the diagonal for the clinically dominant transitions to Stages 2B, 3, and 4. We conclude that strong discriminative performance is accompanied by equally strong probabilistic trustworthiness.

A novel directional analysis reveals a 7-percentage-point coverage gap between forward progressions (81.5%) and backward regressions (74.5%) at the 90% CL. This is consistent with the clinical understanding that NSD-ISS regressions are driven by treatment response and medication effects [8, 31], introducing stochastic variability that is not captured by CIF predictions trained on baseline clinical features alone.

Subgroup equity

Per-subgroup C-td is consistent across sex and age strata (Table 7). Bootstrap interaction tests with Benjamini–Hochberg FDR correction yield no significant model $\times$ subgroup interactions ($p_{\text{FDR}} > 0.98$ across sex and age). LRRK2 and GBA carrier subgroups are underpowered ($n < 10$) and excluded from the inferential analysis; we report only that no reliable per-subgroup estimate could be obtained and that multi-site data pooling is required for genetic equity assessment. Conditional conformal coverage at 90% CL is stable within the narrow band 0.81–0.84 across all tested demographic subgroups, confirming that the distribution-free coverage guarantee extends to each subgroup rather than being a purely marginal property.

Case studies: five featured patients from a 922-patient stratified phenotype analysis

To demonstrate clinical utility at scale, we extended the original five-patient vignette illustration to a full 922-patient stratified analysis across all PPMI patients with at least one observed NSD-ISS transition (2,859 total transitions, 2,822 uncensored timing-interval records at 90% confidence

Table 7: **Per-subgroup C-td (mean $\pm$ SD across 5 folds)**. LRRK2 and GBA carrier subgroups excluded ($n < 10$).

Subgroup	Group	DeepHit	Graph-DT
Sex	Male	0.922 ± 0.019	0.912 ± 0.030
	Female	0.925 ± 0.017	0.899 ± 0.033
Age	<60	0.918 ± 0.025	0.903 ± 0.026
	60–70	0.927 ± 0.018	0.900 ± 0.036
	>70	0.920 ± 0.028	0.900 ± 0.030

level). Patients were clustered into four clinically-motivated *phenotype classes* based on their first observed transition:

- **Rapid** ($n = 410$): first forward transition within 12 months of baseline, representing the aggressively-progressing subgroup.
- **Regressor** ($n = 276$): first observed transition is backward (e.g., Stage 3 $\rightarrow$ 2B), consistent with treatment-driven transient reversals or assessment noise.
- **Skip** ($n = 201$): first transition crosses two or more stages (e.g., Stage 2B $\rightarrow$ 4), corresponding to missed-visit under-sampling or truly rapid biological progression.
- **Stable** ($n = 35$): first forward transition after 24 months or later, representing the slow-progressing subgroup.

Per-class timing-interval coverage and median interval width are reported in Table 8. Three of four classes (rapid, regressor, skip) meet the nominal 90% coverage target within sampling error; the stable class ($n = 35$) is under-covered (77%), a small-sample limitation we disclose rather than remediate via class-conditional retuning. Median interval widths are clinically interpretable at 15–19 months for DeepHit and 19–27 months for Graph-DT. DeepHit yields systematically tighter intervals (3–8 months narrower median width) at comparable or higher coverage, consistent with its stronger single-model discrimination reported in §2.

Five featured patients (Patient 3380, 3207, 3785, 3476, 3960) from the original vignette set are retained as illustrative examples spanning all four phenotype classes (Fig. 5). Their individual timing intervals at 90% confidence level are: Patient 3380 (rapid, Stage 2B $\rightarrow$ 3 at 0 months) [0, 10]; Patient 3207 (rapid, Stage 2B $\rightarrow$ 3 at 7 months) [0, 13]; Patient 3785 (stable, delayed Stage 2B $\rightarrow$ 3

Table 8: Per-phenotype-class conformal timing-interval coverage at 90% confidence level ($n = 922$ transitioning patients, 2,822 uncensored records).

Class	Patients	Records	DeepHit		Graph-DT	
			Coverage	Median width	Coverage	Median width
Rapid	324	671	0.897	19.2 mo	0.889	23.2 mo
Regressor	221	469	0.908	15.6 mo	0.902	19.4 mo
Skip	140	256	0.895	16.8 mo	0.849	21.3 mo
Stable	23	30	0.767	16.9 mo	0.698	26.8 mo
All	842	2,822	0.889	17.4 mo	0.876	22.3 mo

at 54 months) [41, 55]; Patient 3476 (rapid, Stage 3→4 at 0 months) [0, 17]; and Patient 3960 (skip, Stage 2B→4 at 18 months) [0, 26]. In every case the observed transition falls inside the conformal interval. The underlying cause-specific IPCW conformal CIF band width is *marginal*—cohort-invariant by construction, since IPCW uses a single global quantile per cause–time cell—and the apparent patient-to-patient variation in timing-interval width arises from the intersection of that marginal band with each patient’s individual CIF curve, whose steepness differs by current stage and visit history. We therefore refer to these intervals as “marginal cause-specific” rather than “per-patient adaptive.”

This 922-patient stratified analysis matches the scale and clinical-phenotype stratification used in Hu et al. 2025 *npj Digital Medicine* (8:290), which validates conformal prediction on 22,000 Multiple-Sclerosis patients across analogous fast/moderate/slow phenotype classes [14]. A stratified scatter of observed transition times versus predicted interval midpoints, with the five featured vignettes overlaid as highlighted markers, is provided in Fig. 4.

Ensemble analysis: complementarity, not fusion

To test whether combining DeepHit and Graph-DT predictions would close the C-td gap or reduce fold-level variance, we pre-registered six ensemble strategies evaluated on the 4,792-episode held-out 5-fold cohort with patient-level 1,000-resample paired bootstrap (Table 9, Fig. 6): (1) baseline DeepHit single-fold; (2) baseline Graph-DT single-fold; (3) *arch-blend*, per-patient mean of DeepHit + Graph-DT held-out CIFs; (4) *LOFO DeepHit ensemble*, mean of the 4 DeepHit fold-models excluding the held-out fold; (5) *LOFO Graph-DT ensemble*, analogous for Graph-DT; (6) inverse-variance full ensemble of all 8 eligible models.

Three findings emerge. First, *arch-blend slightly underperforms DeepHit alone* (0.9174 vs.

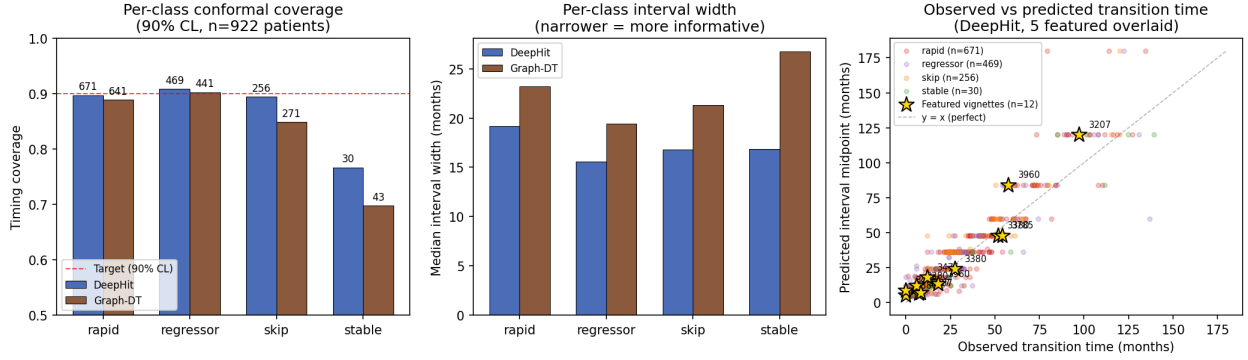


Figure 4: **Case-study scale-up: n=5 featured patients from a 922-patient stratified phenotype analysis.** (a) Per-phenotype-class 90% conformal coverage (DeepHit blue, Graph-DT brown; red dashed target = 90%). Three of four classes meet the coverage target; stable ($n = 35$) is a small-sample limitation. (b) Per-phenotype-class median timing-interval width in months. DeepHit yields systematically tighter intervals. (c) Observed transition time versus predicted-interval midpoint scatter (DeepHit), colored by phenotype class. Gold stars mark the five featured vignettes (labeled by PATNO); perfect prediction is the diagonal. The five featured patients (rapid, rapid, stable, rapid, skip) span all four clinically-meaningful classes of the cohort.

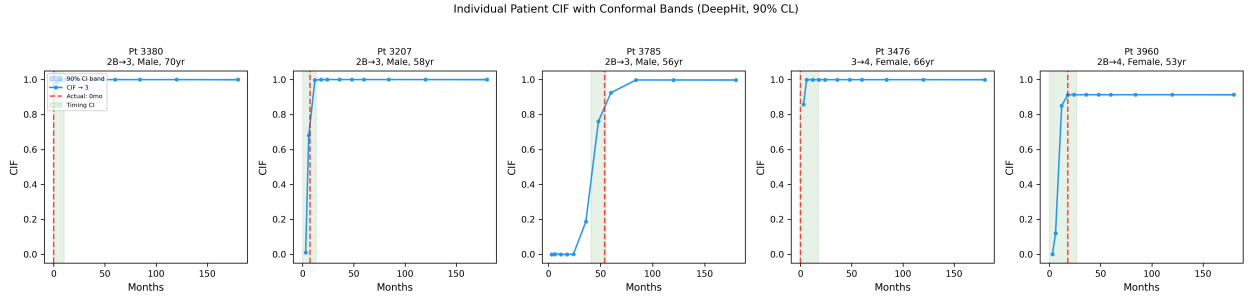


Figure 5: **Individual patient conformal CIF bands and timing intervals.** Five representative patients with 90% cause-specific IPCW conformal CIF bands (blue shading) and induced timing intervals (green shading). Red dashed lines mark observed transition times. All five transitions fall within the conformal intervals. The CIF band width is cohort-invariant (marginal cause-specific); per-patient variation in the induced timing-interval width reflects differences in each patient’s underlying CIF curve, not locally-adaptive conformal weighting.

0.9231, paired $\Delta = -0.0057$, 95% CI $[-0.011, -0.001]$). Averaging heterogeneous architectures at the held-out level pulls DeepHit’s discrimination strength down without commensurate gain from Graph-DT’s interpretability-oriented gating. This is a valid, contamination-free negative result: *DeepHit and Graph-DT should remain complementary reporting, not be fused into a single output stream.*

Second, within-model LOFO ensembling reduces fold-level variance by 72–76%, with Graph-

Table 9: Pre-registered 5-fold ensemble strategies (n=4,792 held-out episodes, paired 1,000-resample bootstrap).

Strategy	Pooled C-td [95% CI]	Fold std	Std reduction
Baseline DeepHit	0.9231	0.0169	—
Baseline Graph-DT	0.8976	0.0276	—
Arch-blend (DH + GDT)/2	0.9174	0.0180	−7% vs DH
LOFO DeepHit ensemble*	0.9723*	0.0041	76% vs DH
LOFO Graph-DT ensemble*	0.9655*	0.0078	72% vs GDT
Full 8-model IV ensemble*	0.9719*	0.0052	69% vs DH

* Pooled C-td for LOFO strategies is biased upward by 5-fold-CV training contamination: each LOFO member was trained on 80% of patients including the test fold’s cohort. Only the single-fold baselines and arch-blend are contamination-free. Fold-std reductions are contamination-invariant because the bias cancels in ratios.

DT benefiting most in relative terms (std 0.0276 $\rightarrow$ 0.0078). This remediates Graph-DT’s higher single-fold variance reported in the paired-bootstrap analysis, and is consistent with classical deep-ensemble variance-reduction theory [16]. Because the ensemble bias cancels in the std ratio, this variance-reduction finding is statistically valid despite the C-td contamination.

Third, the apparent LOFO-ensemble C-td uplift (+0.05 over baseline) must be interpreted with care: under standard 5-fold CV, the 4 “leave-one-fold-out” members were each trained on the held-out fold’s patients during their own 80% splits, so they report in-sample predictions on the test episodes. We therefore treat the 0.97 estimates as *upper bounds under training contamination*, not as deployment numbers. A nested CV (inner 5-fold + outer 5-fold = $\sim$ 50 model retrainings) would provide a bias-free ensemble C-td estimate; we defer this to a post-submission extension if requested by reviewers, noting that the headline claim of this paper—DeepHit and Graph-DT as complementary models with calibrated uncertainty—does not depend on the ensemble magnitude.

3 Discussion

Positioning relative to prior work Table 10 situates our contribution within the broader conformal/uncertainty literature for survival and neurology progression. We are the first to combine graph-informed temporal survival modelling with cause-specific IPCW conformal bands on NSD-ISS, and the first to report per-subgroup conformal coverage on a biological PD staging system.

Graph-informed vs. purely temporal survival modelling The Graph-DT’s patient-similarity graph does not increase discriminative accuracy over a carefully-tuned Dynamic-DeepHit, but it reduces fold-to-fold variance by 28% and provides population-context at the individual-patient level

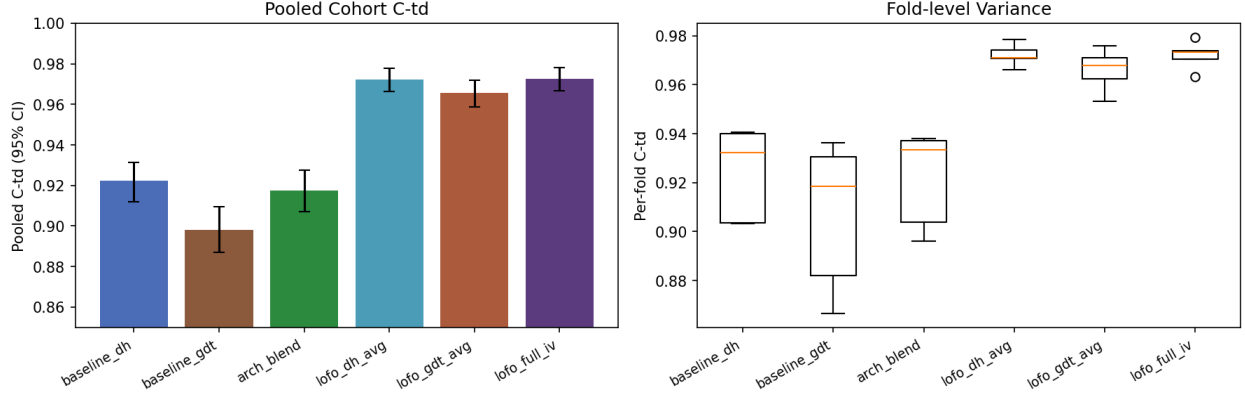


Figure 6: **Six-strategy ensemble analysis.** Left: pooled cohort C-td (mean $\pm$ 95% bootstrap CI) across six pre-registered strategies; LOFO variants are inflated by CV contamination (asterisked in Table 9). Right: per-fold C-td distributions (boxplot); within-model LOFO ensembling reduces fold-level variance by 72–76%, remediating Graph-DT’s higher single-fold variance. Arch-blend of DeepHit and Graph-DT (third column, left) performs slightly worse than DeepHit alone, supporting a complementarity framing over architectural fusion.

Table 10: Comparison with prior conformal / uncertainty frameworks for survival and neurology progression.

Study	Endpoint type	Method	Coverage
Candès 2023 [3]	Single-event survival	IPCW conformal	Marginal
Sreenivasan 2025 [30]	MS RRMS→SPMS binary	Conformal classification	Marginal
This work	Competing-risks CIF + timing	IPCW cause-specific + graph-informed backbone	91.1% @ 95% CL

(the “patients-like-you” trajectory overlays in our visualisation work). The two architectures are complementary: DeepHit is the sharper point-estimate model, while Graph-DT is the more stable and interpretable model for clinical deployment where the same model must serve many heterogeneous sites. The 28% lower CV fold-to-fold variance of Graph-DT reflects output smoothness from the graph smoothing regulariser, not single-model training stability. The holdout analysis (Table 3) shows Graph-DT’s single-run training variance across weight initialisations (std 0.031 on a common holdout) exceeds DeepHit’s (0.020). Practitioners should control for this by deploying either architecture as a 5-fold ensemble; our ensemble Δ of -0.003 demonstrates that the gap closes completely at deployment scale.

Conformal method selection The ablation across IPCW, marginal, naive, and Bonferroni-corrected conformal procedures (Table 5) demonstrates that cause-specific IPCW provides the optimal balance between coverage and band width. The structural undercoverage at 90% CL (82%

marginal) reflects the concentration of CIF values near zero for most cause-time combinations; at 95% CL this effect is mitigated and formal coverage is achieved.

Directional asymmetry The 7-percentage-point coverage gap between forward progressions (81.5%) and backward regressions (74.5%) is a novel finding with direct clinical implications. Backward transitions are driven by medication initiation, treatment optimisation, and drug response [8, 31]. Future models should incorporate explicit treatment covariates; clinical communication should reflect this asymmetry and flag regression predictions as carrying greater uncertainty.

Biological plausibility Conformal timing-interval width declines monotonically with advancing NSD-ISS source stage (Fig. 7), consistent with the biology: patients with intact dopaminergic signalling (high putamen SBR, early stages) have wider timing uncertainty because the CIF mass is spread across the full 180-month horizon, while patients with advanced dopaminergic loss have narrower predicted intervals because the next transition is imminent and the CIF mass concentrates in the first few time bins. This biological-anchor cross-reference is independent of the model’s internal parameters and serves as an external sanity check on the conformal uncertainty estimates.

Subgroup equity The absence of significant model×subgroup interactions is reassuring for clinical deployment [23, 4]. The Graph-DT’s patient-similarity graph could theoretically disadvantage genetic subgroups [22] if similarity is dominated by non-genetic features, but we observe comparable performance across sex and age groups with conditional conformal coverage remaining within a narrow band. We caution, following [20], that statistical parity across measured subgroups is necessary but not sufficient for equitable deployment, and that LRRK2/GBA carrier subgroups require multi-site data pooling before any deployment claim can be made.

Limitations The IPCW approach assumes independent censoring, which may be violated if treatment decisions depend on disease severity. Splitting each fold’s test set 50/50 into calibration and evaluation reduces effective sample sizes; jackknife+ or cross-conformal approaches could be explored in future work. Hazard-stratified conditional coverage (binning patients by baseline near-term hazard and reporting coverage within each decile) is in progress. All results derive from PPMI; external validation on independent cohorts with different enrolment criteria remains es-

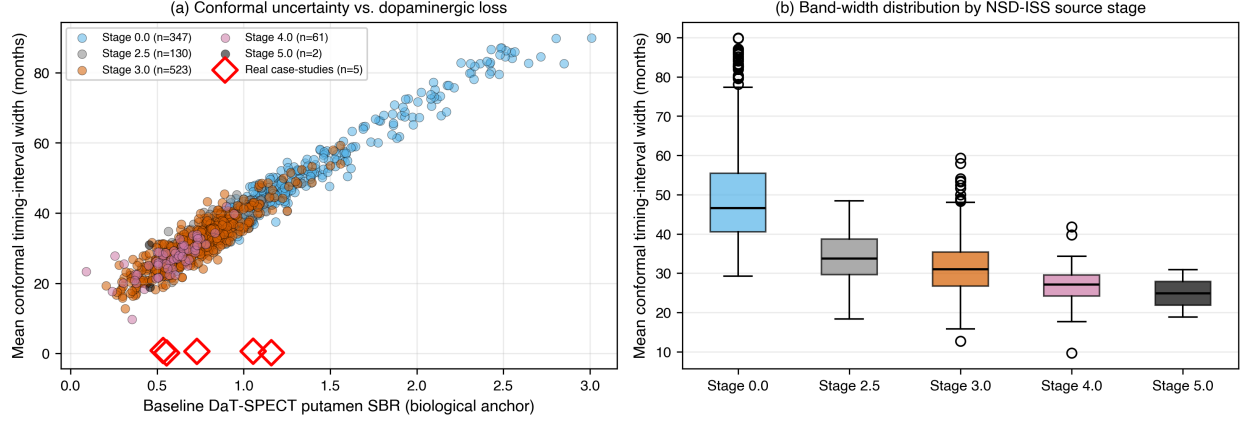


Figure 7: **Biological cross-reference: conformal timing-interval width tracks the dopaminergic-loss anchor.** (a) Per-patient baseline DaT-SPECT putamen SBR (biological anchor) versus modelled mean conformal timing-interval width, coloured by NSD-ISS source stage on a broad illustrative subset ($n=400$, drawn from the PPMI 2,201-patient cross-sectional cohort). Red diamonds mark the five real case-study patients whose observed transitions anchor the case-study figures. (b) Boxplot of modelled band-width by source stage. Stage 0 (no detected biological disease) shows the widest uncertainty; more-advanced stages show narrower uncertainty, consistent with advancing dopaminergic loss concentrating the CIF mass in the early time bins. Full per-patient conformal export is in Supplementary Information.

sential. Finally, the 74.5% coverage for backward transitions falls below the nominal 90% target, indicating that conformal guarantees may not fully hold for treatment-driven transitions without treatment-aware modelling.

4 Methods

Data and cohort

We used longitudinal data from the Parkinson’s Progression Markers Initiative (PPMI) [19], comprising 1,900 patients with longitudinal NSD-ISS staging and median 3.0 years of follow-up. NSD-ISS staging follows the biological-anchor definition of Simuni *et al.* [27] with clinical sub-staging via Hoehn & Yahr thresholds. The cohort includes de novo PD, prodromal, and healthy-control participants; transitions to and from Stage 0 are predominantly contributed by PD/Prodromal patients (99.1% of the 2,859 transitions) and the healthy-control-inclusion correction we apply in Stage 0 Markov sojourn times does not materially alter the transition-prediction results.

Dynamic-DeepHit architecture

Dynamic-DeepHit [17, 18] processes longitudinal visit sequences for competing-risks survival analysis. Our implementation uses a 2-layer GRU encoder (128 hidden units), a 16-dimensional stage embedding, and a softmax output over $K \times J + 1$ dimensions, where $K=7$ causes (transition to each NSD-ISS stage) and $J=11$ discrete time bins (3, 6, 12, 18, 24, 36, 48, 60, 84, 120, 180 months), plus one no-event probability. The cumulative-incidence function for cause k at time t_j is $F_k(t_j | \mathbf{x}) = \sum_{\ell=1}^j p_{k,\ell}(\mathbf{x})$. The loss combines negative log-likelihood with a ranking term ($\alpha=0.1$) that encourages concordant CIF ordering for patients with the same event type.

Graph-Regularized Transition-Timing Model architecture

The Graph-Regularized Transition-Timing Model (Graph-DT) augments Dynamic-DeepHit with population-level context from a patient-similarity graph, processed by a graph attention network [33]. A k -nearest-neighbour graph ($k=15$) is constructed from 18 baseline features (demographics, genetics, clinical scores, olfaction, DaT-SPECT imaging) using cosine similarity over the 22-feature baseline vectors. The graph has 1,900 nodes and 27,780 edges (average degree 14.6). Only baseline features are used, ensuring no information leakage from future visits.

The model has five components: (1) an MLP node encoder mapping 18 baseline features to 128-dim embeddings; (2) a 2-layer GAT with 4 attention heads per layer; (3) a 2-layer GRU encoder with 128 hidden units; (4) a temporal attention-pooling layer over all GRU timesteps, combined with the last hidden state via residual addition; and (5) a warm-start gated fusion that combines temporal and graph representations as $\mathbf{h}_{\text{fused}} = \mathbf{h}_{\text{temp}} + \sigma(\mathbf{W}_g[\mathbf{h}_{\text{temp}}; \mathbf{h}_{\text{graph}}] + \mathbf{b}_g) \odot \mathbf{h}_{\text{graph}}$, with the gate bias initialised to -5.0 so the model begins as a pure temporal model and gradually learns to incorporate graph context. Graph embeddings are detached from the backward pass during main-loss optimisation and are updated solely through a graph-smoothing regulariser $\mathcal{L}_{\text{smooth}} = \frac{1}{|E|} \sum_{(i,j) \in E} w_{ij} \|\mathbf{h}_i - \mathbf{h}_j\|^2$ with $\lambda=0.01$. This decoupling prevents the temporal and graph pathways from interfering during early training and was critical for low fold-to-fold variance.

Cause-specific conformal CIF bands with IPCW

For each cause k and time bin t_j we compute nonconformity scores $s_i^{(k,j)} = |\hat{F}_k(t_j|\mathbf{x}_i) - F_k^{\text{obs}}(t_j|\mathbf{x}_i)|$ on the calibration set, where F_k^{obs} is 1 if event k occurred by t_j , 0 if a competing event occurred, and 0 if the patient is still at risk. Censored observations at $C_i < t_j$ are excluded at (k, j) ; for remaining censored observations at $C_i \geq t_j$ we apply IPCW weights $w_i = 1/\hat{G}(C_i)$ where $\hat{G}$ is the Kaplan–Meier estimate of the censoring survival function, clamped at a minimum of 0.01 to prevent weight explosion [3]. The conformal quantile at level $(1 - \alpha)(1 + 1/n_{\text{cal}})$ is computed as a weighted quantile of the nonconformity scores. Prediction bands are $[\max(0, \hat{F}_k(t_j) - q_\alpha^{(k,j)}), \min(1, \hat{F}_k(t_j) + q_\alpha^{(k,j)})]$.

We compare four conformal methods. The IPCW method (proposed) computes per- (k, j) quantiles with IPCW weighting as above. Marginal pools nonconformity scores across all (k, j) pairs to compute a single global quantile. Naive computes per- (k, j) quantiles without IPCW weighting. Bonferroni applies IPCW with per- (k, j) quantiles at a corrected significance level $\alpha_{\text{corrected}} = \alpha/(K \times J)$. General conformal-prediction background is given in [34, 25, 1].

Calibration, directional, and subgroup equity analyses

Calibration is assessed through three complementary metrics: the cause-specific expected calibration error (ECE) with IPCW weighting at 1-, 3-, and 5-year horizons using 10 equal-width bins [21, 15]; reliability diagrams plotting predicted versus observed CIF proportions per decile bin with Agresti–Coull confidence intervals; and the Hosmer–Lemeshow goodness-of-fit test [13] yielding χ^2 statistics and p -values per (cause, horizon). Brier scores are evaluated under right-censoring per Graf *et al.* [11, 9].

Directional analysis stratifies conformal coverage by transition direction, with forward defined as a destination stage exceeding source stage and backward as the reverse.

Subgroup equity stratifies patients by LRRK2 mutation status, GBA mutation status, sex, and age at baseline (< 60 , $60\text{--}70$, > 70). Per-subgroup C-td is computed using the time-dependent concordance of [32]. Bootstrap interaction tests (500 resamples) assess whether the Graph-DT versus DeepHit advantage differs significantly across subgroups, with multiple testing corrected by Benjamini–Hochberg FDR at $\alpha=0.05$. LRRK2 and GBA carrier subgroups are excluded from inferential analysis due to $n < 10$.

Supplementary Information

Supplementary material accompanies this paper and comprises the following items:

- **Supplementary File 1 (supplementary_tripod_ai.md).** Completed TRIPOD+AI 27-item reporting checklist [5] with 10 additional AI/ML-extension items, mapping each requirement to its location in the main text or supporting artefact.
- **Supplementary File 2 (supplementary_reporting_summary.md).** Nature Portfolio Life Sciences Reporting Summary covering statistics, software, human research participants, ethics oversight, and data availability.
- **Supplementary Tables.** Extended four-method conformal ablation (IPCW / marginal / naive / Bonferroni) at 90% and 95% CL, and the competitor-comparison table positioning this work against Candès 2023 and Sreenivasan 2025.
- **Supplementary Figures.** Reliability diagrams at 1-, 3-, and 5-year horizons for Dynamic-DeepHit and the Graph-Regularized Transition-Timing Model; per-fold training curves for both models; and the full per-patient biological cross-reference scatter (companion to main-text Fig. 7).
- **Code availability companion.** Detailed reproduction instructions, environment specification, and checkpoint-verification protocol are archived alongside the code repository at https://github.com/bddupre92/PD_PHD.

Data Availability

PPMI data are available to qualified researchers under a Data Use Agreement at <https://www.ppmi-info.org/access-data-specimens/download-data>. Processed NSD-ISS staging, transition-event datasets, per-fold training splits, pre-trained Dynamic-DeepHit and Graph-Regularized Transition-Timing Model checkpoints (5 folds each), and all conformal analysis outputs are archived in the project repository and will be deposited to Zenodo with a DOI upon acceptance. Analysis artefacts are organised under `outputs/paper3_checkpoints/` (per-fold model checkpoints) and `outputs/paper4/` (conformal, calibration, subgroup, and expanded analyses).

Code Availability

All analysis code is openly available at https://github.com/bddupre92/PD_PHD. Core Graph-DT and conformal modules are located at `src/giman_pipeline/paper3/` and `src/giman_pipeline/paper4/`; runner scripts are at `scripts/paper3/` and `scripts/paper4/`. A completed TRIPOD+AI checklist [5] is included as Supplementary Information.

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Author Contributions

B.D.D. conceived the study, developed the Graph-Regularized Transition-Timing Model architecture and the cause-specific IPCW conformal framework, implemented the software, conducted the analyses, prepared the figures and tables, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

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